

What I'm going to talk about to you today is something called tea tree oil, which was first extracted from *Melaleuca alternifolia*. In Australia, this species remains the most important commercially. Several other species are cultivated for their oil extraction. There is a very long history of tea tree oils use in aroma therapy. Traditionally, *Melaleuca alternifolia* leaves were crushed, and the oil was inhaled by the aborigines of Australia for the treatment of coughs, colds and also for the treatment of wounds. For instance, they chewed the young leaves to alleviate headaches and took them to treat sore throat or skin ailments.

The aborigines world was discovered by William Lee and soon a Dutch explorer, who was the first European to sell to Australia in 1606, he reached the northern coast of Australia in his ship. Then several voyages of exploration followed in the first half of the 17th century. The Dutch founded a paradise on earth for man's wellbeing with timber, stone and lime for building. There was also plenty of salt and the coast was full of fish. Besides, they found the characteristics of the diet there because they happened to meet ten naked black aborigines, having a meal in the open air.

While the value of tea tree oil originated from Australia. It was gradually known and tested by the outsiders in the middle of the 18th century. So Hugh palace, an officer of the British Royal Navy, who had been to Australia several times during that period, got serious injuries all over due to his experiences in several wars. For more than the last 15 or 16 years of his life, he seldom laid down in a bed because of the constant pain in his leg. Then he tried tea tree oil, as it was said that tea tree oil could operate as a very powerful immune stimulant for pre and post surgical care.

The use of the name tea tree also called paper bark trees probably originated from captain James cooks description he made soon after he had arrived at the coast of New South Wales in 1770. At the time he witnessed some aborigines of Australia using one of the shrubs leaves to make an infused drink in place of tea.

In the 1920s, some human clinical research and the documentation of many benefits associated with tea tree oil were credited, which were made by doctor Arthur penfold on Australian government chemist. He investigated the business potential of a number of native extracted oils, then reported that tea tree oil was promise ink as it existed powerful antiseptic properties. But after World War Two, the entry of antibiotics declined the use of natural products in medicine, which had a negative effect on the production of tea tree oil as such an important and valuable material in the world.

How is tea tree oil produced? I think most of you are curious about this tea tree oil can be extracted in some different ways, but the most traditional way is steam distillation.

Once harvested in winter, when the amount of required essence in oil meet the needs for production. The finally cut trees are transported to a steam distillation facility. The extraction is made by distilling the leaves in specially designed stainless steel stills,

along with the stems to yield pure oil. The water filled boiler is heated and constantly monitored to maintain the correct temperature. Both the steam and oily evaporate and then condense as they run through a pipe into the collecting container, where the oil floats to the lid. While the water, because of gravity go steadily out the lower exit pipe, at the end of the hour, the oil is siphoned off through the upper pipe, while the condensed steam floats through the lower pipe, towards the ground. At the end of each distillation, all the spent plant material is hauled out of the still pot. By hand, with a short rake, piled onto a trailer and spread where required as a thick woody mulch.